METRIC

DOD-D-51467(EA)
AMENDMENT 3
10 February 1988
SUPERSEDING
AMENDMENT 2
4 June 1986

MILITARY SPECIFICATION

DECONTAMINATING SOLUTIONS (METRIC)

This amendment forms a part of DOD-D-51467(EA), dated 25 February 1980, and is approved for use within the Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

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Add immediately after paragraph 2.2 the following reference:

"Reagent Chemicals, American Chemical Society Specifications, Sixth Edition, 1981.

"(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth Street N.W., Washington, DC 20036.)"

Delete paragraph 3.2.1 in its entirety and substitute the following:

"3.2.1 Ethyl alcohol, undenatured, ACS grade. The ethyl alcohol content shall be 72 ± 2 percent by weight when tested as specified in 4.2.4.1."

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Delete paragraph 3.3.1 in its entirety and substitute the following:

"3.3.1 Ethyl alcohol, undenatured, ACS grade. The ethyl alcohol content shall be 45 ± 2 percent by weight when tested as specified in 4.2.4.1."

AMSC N/A

FSC 4230

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4.2.4.4, line 4: Delete "0.1N" and substitute "1.0N".

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4.2.4.5: Delete in its entirety and substitute the following:

"4.2.4.5 Zinc chloride content. Determine zinc chloride content by one of the following methods:

4.2.4.5.1 Weigh 5 g of sample to the nearest milligram. Dilute to 1 liter. Dilute 5 ml of this solution to 500 ml and determine the weight of zinc chloride on an atomic absorption unit using a zinc cathode lamp comparing the results to a standard curve obtained for known zinc chloride solutions run at the same time. Calculate the percent by weight zinc chloride in the solution as follows:

$$\text{Percent zinc chloride} = \frac{100AB}{W}$$

where A = Weight of zinc chloride from calibration curve in grams
B = Dilution factor, and
W = Weight of sample in grams

4.2.4.5.2 Weigh to the nearest 0.1 milligram 5.0 grams of sample into a 250 ml flask. Add 100 ml of distilled water. Add concentrated ammonium hydroxide, drop-wise, until zinc begins to precipitate (approximately 5-6 drops will be required). Pipet 5 ml of pH 10 buffer solution into the flask. The sample solution should now be clear. Titrate with 0.05M Ethylene Diamine Tetra-acetate (EDTA), using Eriochrome Black T as an indicator. The color change shall be from purple to deep blue at the end point. Calculate the percent by weight zinc chloride in the solution as follows:

$$\text{Percent zinc chloride} = \frac{13.629AB}{W}$$

where A = Molarity of EDTA solution
B = ml of EDTA solution
W = Weight of sample in grams"

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